

BLACKBOX X — Strategy Genome

Quantitative Relationship Network Reference Manual — The
Metaphor of the "Genome" — Central Research Question — Data
Contracts

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SYSTEM

BLACKBOX X Engine

STATUS

Documentation

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BLACKBOX X — Strategy Genome

Quantitative Relationship Network Reference Manual

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TypeScript Verification: Clean compile (`npx tsc --noEmit`) — 0 errors)

Bundle Target: Production Vite Build (`npm run build`) — 0 errors)

1. Concept

Strategy Genome is an interactive quantitative relationship network that maps the structural topology connecting:

Assets ↔ Correlations ↔ Volatility ↔ Strategies ↔ Market Regimes ↔ Stress Responses

The Metaphor of the "Genome"

In genomics, a genome represents the complete set of structural instructions and connections that define how an organism expresses traits under varied environmental stimuli. In BLACKBOX X, the **Strategy Genome** serves as the **structural fingerprint of the quantitative market and strategy environment**.

IMPORTANT

Strategy Genome is **not** a biological simulation, nor is it a decorative neon force-directed graph. Every node radius, edge weight, line dash pattern, coordinate vector, and inspector metric corresponds strictly to deterministic, calibrated mathematical output from the verified BLACKBOX X engines.

Central Research Question

The system is designed to answer:

"How does the quantitative structure of this market and tested trading strategies transform across baseline, regime-shifted, and stressed market states?"

2. Graph Model & Topology

The graph engine is implemented in `src/core/strategyGenome.ts` with strict TypeScript types and zero `any` usage.

Data Contracts

Node Definition (**GenomeNode**)

```
export interface GenomeNode {
  id: string;
  type: GenomeNodeType; // 'ASSET' | 'STRATEGY' | 'REGIME' | 'RISK_METRIC'
  label: string;
  sublabel?: string;
  value: number;
  normalizedValue: number; // [0, 1] for visual radius mapping
  x: number; // Normalized coordinate [-1, 1]
  y: number; // Normalized coordinate [-1, 1]
  radius: number; // Viewbox radius [px]
  color: string;
  metadata: {
    asset?: Asset;
    strategy?: StrategyType;
    regime?: RegimeType;
    metrics?: Record<string, number | string>;
    stressed?: boolean;
  };
}
```

Edge Definition (**GenomeEdge**)

```
export interface GenomeEdge {
  id: string;
  source: string;
  target: string;
  value: number; // Quantitative relationship value (e.g. Pearson correlation)
  normalizedStrength: number; // [0, 1] for stroke width scaling
  relationship: 'CORRELATION' | 'APPLICATION' | 'REGIME_EXPOSURE' | 'STRESS_SHIFT';
  isNegative?: boolean; // Controls dashed rendering for negative correlation
  metadata: {
    label: string;
    description: string;
    window?: number;
    mode?: 'STATIC' | 'ROLLING';
    baselineValue?: number;
    stressValue?: number;
  };
}
```

Snapshot Definition (**GenomeSnapshot**)

```
export interface GenomeSnapshot {
  nodes: GenomeNode[];
  edges: GenomeEdge[];
  mode: GenomeMode; // 'MARKET' | 'STRATEGY' | 'REGIME' | 'STRESS'
  selectedAsset: Asset;
  selectedStrategy: StrategyType;
  fingerprint: StrategyRiskFingerprint;
  summary: {
    nodeCount: number;
    edgeCount: number;
    meanCorrelation: number;
    dominantRegime: string;
    activeAssetVol: number;
  };
}
```

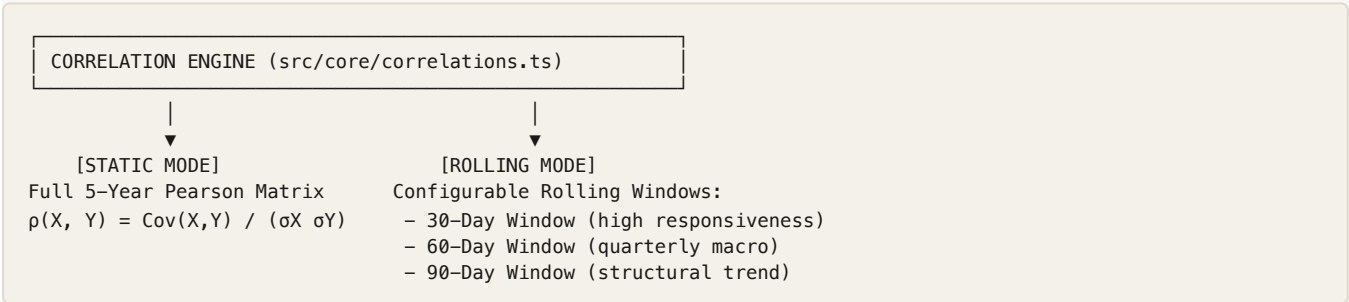
3. Data Mapping & Quantitative Integrity

Every visual property maps directly to real calculations from the primary math engines:

Visual Attribute	Target Element	Mathematical Formula / Source	Range / Scaling
Asset Node Radius	ASSET Nodes	Normalized Annualized Volatility σ_{ann} from <code>metrics.ts</code>	$r \in [24\text{px}, 42\text{px}]$
Strategy Node Radius	STRATEGY Nodes	Normalized Sharpe Ratio S from <code>backtest.ts</code>	$r \in [22\text{px}, 36\text{px}]$
Regime Node Radius	REGIME Nodes	Sub-period duration % or Win Rate from <code>regimes.ts</code>	$r \in [20\text{px}, 32\text{px}]$
Edge Stroke Width	Correlation Edges	Absolute Pearson Correlation	$ \rho $
Edge Dash Style	Negative Correlations	<code>stroke-dasharray="6,4"</code> if $\rho < 0$, else solid	Binary structural distinction
Node Color	Asset / Strategy	Asset token colors (GOLD : <code>#E5A93C</code> , BTC : <code>#F7931A</code> , NVDA : <code>#76B900</code>)	Standard platform palette

4. Correlation Methodology

Static vs. Rolling Mode



1. **Static Mode:** Evaluates daily log returns over the entire 5-year historical horizon (1,826 observations per asset).
2. **Rolling Mode:** Evaluates rolling Pearson correlation over lookbacks of 30, 60, or 90 calendar days:

$$\rho_t(X, Y) = \frac{\sum_{i=0}^{W-1} (R_{X,t-i} - \bar{R}_X)(R_{Y,t-i} - \bar{R}_Y)}{(W - 1) s_{X,t} s_{Y,t}}$$

The active edge thickness transitions dynamically to reflect the latest point of the selected rolling series.

Non-Causal Semantics Rule

- **Allowed Descriptive Terms:** "displays linear correlation", "exhibits co-movement", "correlation weakened", "relationship inverted".
- **Strictly Prohibited Causal Terms:** "causes", "drives", "predicts", "leads to", "triggers".

5. Strategy Mapping & Comparison Mode

Strategy Genome projects the backtest outcome of trading models without establishing an artificial "winner".

Supported Strategy Types

- **SMA Crossover:** Trend-following via moving average crossovers (Fast Lookback: 10–30d, Slow Lookback: 40–100d).
- **EMA Trend:** Exponential smoothing responding to directional momentum while mitigating lag.
- **Momentum:** Relative rate-of-change over lookback horizon with directional threshold filter.
- **Mean Reversion:** Contrarian positioning based on statistical deviation (Z-score / Bollinger bands) around historical mean.

Objective Comparative Inspection

When in **STRATEGY Mode**, all four strategies are arrayed around the selected asset. The inspector displays:

- **Total Return (%)**
- **Sharpe Ratio (S)**
- **Annualized Volatility (%)**
- **Maximum Drawdown (%)**
- **Trade Count (N_{trades})**

NOTE

BLACKBOX X does not tag any strategy as "BEST", "TOP PERFORMER", or "WINNER". Strategies display differing risk-return profiles depending on market conditions, trade frequency, and asset volatility.

6. Regime Mapping

In **REGIME Mode**, the graph connects the active strategy to the four classified market environments detected by `src/core/regimes.ts`:

1. **BULL REGIME:** Positive 60-day returns with moderate annualized volatility.
2. **BEAR REGIME:** Negative 60-day returns with elevated volatility.
3. **HIGH VOLATILITY REGIME:** Annualized volatility above the historical 75th percentile.
4. **LOW VOLATILITY REGIME:** Compressed volatility below the historical 25th percentile.

Metric Attribution

Each regime node exposes real sub-period performance:

- **Regime Total Return:** Equity growth achieved exclusively while in that regime.
- **Regime Sharpe Ratio:** Risk-adjusted excess return normalized to sub-period duration.
- **Regime Win Rate:** Percentage of trades closed profitably during that market regime.

7. Stress Mapping (Time-Travel Macro Shock Lab Integration)

When **STRESS Mode** is active, Strategy Genome directly ingests the current `stressResult` produced by `src/core/stressTesting.ts`:

Dual Baseline vs. Stressed Representation

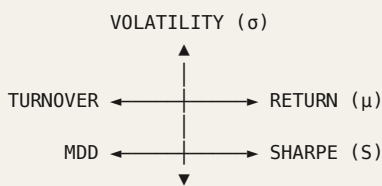
- **Asset Nodes:** Shows baseline historical volatility alongside stressed volatility:

$$\sigma_{\text{stressed}} = \sigma_{\text{baseline}} \cdot M_{\text{vol}}$$

- **Drawdown Comparison:** Directly displays baseline maximum drawdown vs. crisis drawdown (e.g. -18.4% baseline vs -41.2% under COVID-19 liquidity shock).
- **Crisis Correlation Convergence:** Pairwise edges update to reflect correlation measured *during the crisis window*:
 - In liquidity crises, cross-asset correlations often surge toward $+1.0$ as investors liquidate liquid assets simultaneously.
 - The graph renders stressed edges with thicker stroke widths and red/amber stress accents to highlight diversification breakdown.

8. Strategy Risk Fingerprint (5D Radar Architecture)

The **Strategy Risk Fingerprint** renders a normalized 5-axis polygon representing the multidimensional risk profile of the active strategy:



Normalization Scale

Each axis maps actual values into $[0, 1]$ based on empirical multi-asset boundaries:

- **Return Axis:** Normalized over $[-20\%, +60\%]$
- **Volatility Axis:** Normalized over $[10\%, 80\%]$
- **Sharpe Axis:** Normalized over $[-0.5, +2.5]$
- **Drawdown Axis:** Normalized inverted scale where smaller drawdown yields higher score
- **Turnover Axis:** Normalized over $[0, 100 \text{ trades}]$

Actual quantitative numbers are displayed directly in the badge chips alongside the diagram.

9. Ghost Mode 2.0 Integration

Strategy Genome is directly wired to the **Ghost Mode 2.0 Research Trail**:

- **Insight Category:** `GENOME_RELATIONSHIP_SHIFT`
- **Trigger:** Detected when rolling correlation shifts by $|\Delta \rho| \geq 0.15$ over 60-day windows.
- **Actionable Next Test:** Dispatches `NAVIGATE` actions with target `'genome'` or `'stress'`, enabling seamless hypothesis-driven navigation between correlation analysis, genome topology, and stress testing.

10. Performance & Accessibility Safeguards

Performance Optimizations

- `useMemo` **Memoization:** Graph layout, node positioning, edge calculations, and 5D fingerprint coordinates are wrapped in `useMemo` blocks keyed strictly to `[prices, asset, strategy, params, mode, correlationMode, correlationWindow, stressResult]`.
- **Zero Calculation on Hover:** Hovering or clicking nodes/edges performs constant-time state lookups ($O(1)$) without re-running backtests or correlation matrices.

- **SVG Hardware Acceleration:** The graph is rendered with vector SVG paths utilizing CSS transforms and Framer Motion spring physics.

Accessibility & Reduced Motion

- **Reduced Motion Support:** Inspects `useReducedMotion()` from Framer Motion. Transitions collapse from animated springs to zero-duration opacity changes.
- **Multi-Channel Encodings:** Negative correlations utilize dashed stroke patterns (`stroke-dasharray="6,4"`) in addition to color distinctions to ensure full readability for color-blind researchers.
- **Semantic ARIA Badging:** Controls, SVG nodes, and inspector panels feature standard semantic roles and descriptive titles.

11. Interpretation Limitations

1. **Non-Predictive Historical Representation:** Strategy Genome illustrates structural relationships within historical data (2019–2023) or simulated stress models. It does not predict future price trajectories or structural breaks.
2. **Linear Metric Dependence:** Correlation edges utilize the Pearson product-moment correlation coefficient, which quantifies linear co-dependence. Non-linear coupling, tail dependence, or asymmetric downside copulas are not captured by single-point correlation values.
3. **Execution Assumptions:** Backtest metrics shown on strategy nodes assume next-bar close fills with 0.10% round-trip transaction costs and zero intraday slippage or liquidity constraints.